

CE807 - Report

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Abstract

The present work provides an analysis of two different methods of classifying text into one of two possible classes, namely a supervised Multilayer Perceptron (MLP) and a zero-shot FLAN-T5-small language model. While the MLP utilised TF-IDF representations with features consisting of unigrams and bigrams, the FLAN-T5-small performed classification based on five different prompts in natural language. Both methods were evaluated in terms of their macro F1-score calculated on a separate validation set. The MLP yielded an F1-score of 0.76, whereas the best F1-score for the FLAN-T5-small was 0.848 achieved with Prompt 2. The results show that the zero-shot LLM surpassed the conventional supervised neural network method, demonstrating the usefulness of contextual representations and instruction tuning.

Materials

- [Google Colab Code](#) (Mandatory)
- [Zoom Recoded Presentation](#) (Mandatory)
- [Google Drive Folder](#) containing models and saved outputs

1 Task 1: Model Discussion

1.1 Summary of Models

Multilayer Perceptron (MLP): Multilayer Perceptron (MLP) was chosen as the supervised learning algorithm for sentiment analysis. Prior to sentiment classification, the text reviews were represented by numeric features using the Term Frequency–Inverse Document Frequency (TF-IDF) method and taking into account both unigrams and bigrams. MLP is a feed-forward artificial neural network comprising interconnected neurons organized into input, hidden, and output layers. Contrary to linear classification algorithms, an MLP, with the help of the hidden layer and non-linear

activation function, is capable of learning more complicated relationships between textual features and sentiment classes. In the current research, the MLP was trained using the labeled dataset of reviews in order to classify them into the positive and negative sentiment categories. Then, the validation dataset was used to compute the macro F1-score of the model. The MLP was selected since the neural networks can learn non-linear decision boundaries, and the TF-IDF represents the words and phrases efficiently.

Zero-Shot LLM (FLAN-T5-small): The next model chosen for sentiment analysis was FLAN-T5-small – an open-source instruction-tuned language model that works with zero-shot prompting. Contrary to the MLP, the FLAN-T5-small model was not trained on the particular sentiment dataset used in this work. On the contrary, each review was embedded into a natural-language instruction that asks the model to classify it as positive or negative. For this reason, five different prompts were implemented in order to evaluate whether the different wording of instruction influences the classification results (Wang et al., 2023). After that, the produced output was post-processed into binary sentiment classes (positive is equal to 1; negative is equal to 0). The macro F1-score was computed for each prompt with the help of the validation dataset. It helps to select the best performing prompt formulation. FLAN-T5 can be used for such purposes as instruction tuning improves language model's ability to generalise to out-of-domain tasks and process natural language instructions without re-training (Chung et al., 2024). Longpre et al. (2023) also provide additional evidence on the efficiency of FLAN instruction tuning (Longpre et al., 2023).

1.2 Critical discussion and justification

Justification of MLP Model: The MLP was chosen for being the supervised model due to its ability to discover nonlinear mappings between TF-IDF

features and sentiment categories. As opposed to a linear classifier, hidden layers and activation functions of the MLP allow for the creation of more complex decision boundaries, making it appropriate for distinguishing positive and negative textual patterns. TF-IDF is especially good at putting emphasis on informative terms, diminishing the impact of frequently used words (Kowsari et al., 2019). This method, however, has certain drawbacks since TF-IDF treats words as independent elements, and is thus unable to detect semantic similarity or contextual meanings. For this reason, it might be challenging for MLP to process the textual patterns with sarcasm, ambiguity or context. There is recent evidence suggesting that pre-trained transformer models perform better than the regular neural networks based on TF-IDF features. MLP, nevertheless, serves as a good computationally efficient supervised baseline model in contrast to the zero-shot LLM.

Critical Analysis of MLP Model: MLP parameters were set up in a way to balance the ability to perform classification, speed and avoid overfitting. The TF-IDF algorithm has the maximum number of features equal to 6,000. This helps in limiting dimensionality while preserving valuable vocabulary. Unigrams and bigrams can be expressed by setting the n-gram range equal to (1,2). This helps express both individual terms and two-term combinations that may contain the emotional meaning of texts. English stop-word removal filters out frequent words which are not good at discrimination. The MLP has only one hidden layer with 100 neurons. This ensures enough flexibility to learn nonlinear dependencies without adding complexity. ReLU was chosen as the activation function, while Adam solver ensured efficient optimization. The number of iterations was limited by the maximum of 100 to prevent too many runs, and random state provided the opportunity of reproducibility. Finally, early stopping is performed to terminate training when validation performance stops increasing. It is a well-known technique to improve generalization of neural networks (Prechelt, 1998).

Justification of Zero-Shot LLM: Model 2 was chosen to be FLAN-T5-small because of the nature of its instruction-tuned architecture which makes it capable of zero-shot classification without needing further training for each task. This is due to the interpretation of natural language instructions by the model to generate a sentiment class. FLAN-

T5-small is inherently very different from the supervised MLP since it uses instructions to classify sentiments. There is proof of the benefits of instruction tuning in improving the generalisation ability to unseen tasks and in zero-shot learning (Chung et al., 2024). Nonetheless, zero-shot classification will always be susceptible to prompting, the nature of the task and the size of the model itself. Therefore, FLAN-T5-small might provide inconsistent results for ambiguous, sarcastic, and context-dependent prompts. Also, its small size means it has limited reasoning capabilities when compared to other language models.

Critical Analysis of Zero-Shot LLM: FLAN-T5-small was utilised with zero-shot prompting where sentiment examples were not provided to the model before sentiment classification. This approach was used to test how useful instruction tuning would be in identifying sentiments as positive and negative. Five prompts were constructed in order to vary them in wording and degree of specificity since prompting plays an important role in zero-shot classification. The response from the model was constrained by setting max new tokens=5 since the generated sentiment should be a single word or label. do sample=False was utilised in order to generate deterministic output. The input text was also truncated up to 512 tokens since larger amounts might result in more computation resources usage. The output sentiment was transformed into binary form – 1 for positive and 0 for negative sentiments. This is a straightforward way of performing zero-shot classification. Previous studies have shown that natural-language instructions are crucial for zero-shot generalisation, and prompt formulation variety helps instruction-tuned models.

2 Task 2: Design, implementation of classifiers

2.1 Model Implementation Details

Multilayer perceptron (MLP) Model: Multilayer perceptron (MLP) classifier was coded in Python utilising Scikit-learn. Initially, training and validation sets were loaded with Pandas, and missing text values were replaced by empty strings. Labels with positive sentiments were mapped into 1, while negative sentiments became 0. Text was represented numerically by using TfidfVectorizer with the maximum number of 6000 features and unigrams and bigrams. The extracted features were applied to MLPClassifier without any additional preprocess-

ing through Scikit-learn pipeline. One hidden layer with 100 neurons was included in the model with an early stopping technique to prevent overfitting. Scikit-learn’s MLP classifier uses iteration to optimise classification loss and provides early stopping option based on validation results. The model was trained on the training data and validated on the validation dataset in terms of F1-score. The final model was saved as `mlp_model.pkl` using Joblib and then loaded to predict `test.csv`.

Zero-Shot LLM Implementation Details: The zero-shot classifier was built using the Google/Flan-T5-small open-source Hugging Face model. FLAN-T5 is an extension of the T5 model and it is pre-trained on instructions; therefore, it does not require any further fine-tuning in order to be applied to a new task. The tokenizer and the model itself were loaded using `AutoTokenizer` and `AutoModelForSeq2SeqLM` respectively, and were moved to the GPU if CUDA was available. Each input was shortened to limit the computation costs and added to five different natural language prompt templates. Each prompt asked the model to determine if the text was positive or negative in its sentiment. The tokenized prompts were fed to the model, and the output was generated using the `model.generate()` method with deterministic decoding. The output was lowered to lowercase and transformed to binary values – 1 for positive and 0 for negative sentiments.

Dataset	Total	% Class A	% Class B
Train	4914	79.60	20.40
Valid	700	79.57	20.43
Test	1386	80.1	19.9

Table 1: Dataset Details

The distribution of observations in each dataset can be seen in Table 1. The training dataset consists of 4,914 observations, being the biggest subset of the data, while validation and test datasets consist of 700 and 1,386 observations, correspondingly. Distributions of classes among subsets are almost identical in all three sets. Class A makes up 79.60 percent of observations in the training set compared to 20.40 percent in Class B. Class A makes up 79.57 percent of observations in the validation set compared to 20.43 percent in Class B. Finally, the test set includes 80.1 percent and 19.9 percent of Class A and Class B observations correspondingly. This means that the original class distribution has

been correctly maintained in dataset subsets. Nevertheless, this also means that the results include obvious class imbalance since Class A occurs approximately four times more frequently than Class B. Thus, macro F1-score would be appropriate to estimate model performance.

Model	F1 Score
Multilayer Perceptron (MLP)	0.76
Zero-Shot LLM	0.848(prompt 2)
SoTA (Devlin et al., 2019)	0.95

Table 2: Model Performance

Table 2 shows the comparison between the performance of the MLP, Zero-Shot LLM (FLAN-T5-small) and the state-of-the-art (SoTA) BERT model in terms of classification accuracy based on the F1-score. MLP demonstrated a good F1-score of 0.76 which implies its reasonable classification ability although the model had some problems with properly distinguishing the two classes. In contrast, FLAN-T5-small had a better F1-score of 0.848 when used with Prompt 2. It means that instruction-tuned language modelling and zero-shot prompting had proven to be more efficient than the MLP based on TF-IDF for the present classification problem. Among the tested prompts, Prompt 2 gave the best result for the FLAN-T5-small. Nevertheless, both applied models still did not outperform the referenced SoTA BERT (0.95 (Devlin et al., 2019)).

Table 3: Hyperparameters Used for the MLP and FLAN-T5-small Models

Model	Hyperparameter	Value
MLP	TF-IDF Max Features	6000
MLP	N-gram Range	(1, 2)
MLP	Stop Words	English
MLP	Hidden Layer Size	100
MLP	Maximum Iterations	100
MLP	Early Stopping	True
MLP	Random State	2501073
FLAN-T5-small	Maximum Input Length	512
FLAN-T5-small	Maximum New Tokens	5
FLAN-T5-small	Sampling	False
FLAN-T5-small	Number of Prompts	5
FLAN-T5-small	Classification Type	Zero-shot

Hyperparameters of the MLP and FLAN-T5-small models have been discussed in Table 3. Regarding MLP, TF-IDF vectorizer had been allowed to keep 6,000 terms consisting of uni- and bi-grams

as features. The number of neurons in the hidden layer had been set to 100 and only 100 iterations had been performed during training using early stopping technique. Random state had been kept constant to make results reproducible. As for FLAN-T5-small, the input text had been restricted to 512 tokens, and response generated had been limited to five tokens since only short sentiment labels were needed. Sampling had been switched off in order to have deterministically reproducible predictions. The impact of different formulations of zero-shot prompts on classification results had been studied by evaluating five such prompts.

3 Task 3: Output Comparison and Analysis

3.1 Justification of Models Performance

Multilayer Perceptron (MLP): An F1-score of 0.76 was obtained by the MLP, which is a sign of acceptable performance but lesser effectiveness compared to the zero-shot LLM. The performance of the MLP is mainly credited to the TF-IDF representation, which successfully identifies statistically significant features as either unigrams or bigrams of the two classes. Another element contributing to the performance of the MLP is the hidden layer of 100 neurons, which helps in learning the non-linear relationship among these features. On the other hand, the TF-IDF representation is not based on the meaning of the text but rather on the occurrence and significance of terms in the text. Thus, the MLP would have a challenge in classifying text whose classification is based on complicated sentence structure, inference, ambiguity, and context-dependent meanings. In addition, class imbalance in the dataset could be a contributing factor due to the fact that almost 80 percent of the samples fall into Class A.

Zero-Shot LLM (FLAN-T5-small): FLAN-T5-small performed better, receiving the higher F1-score of 0.848 using Prompt 2 and outperforming MLP by 0.088. It can be attributed to the pre-trained language representations and instruction tuning available in FLAN-T5. Unlike the TF-IDF, the language model understands words within the context, being able to analyse the relationship between the words and the meaning of the input altogether. Moreover, thanks to the instruction tuning ability, FLAN-T5 can execute new classification tasks based on the natural-language prompts without having any extra task-specific training. Prompt

2, which contained the question on the sentiment’s polarity directly, gave the best performance, implying that an explicit instruction helped understand the task at hand. Nevertheless, the F1-score was still lower than the one in the SoTA, showing certain limitations connected to the relatively small model size and zero-shot approach.

3.2 Comparison of Models with Diverse Examples

Diverse examples-based comparison between the MLP and FLAN-T5-small has been done to reveal the difference between the performance of the two classifiers other than their overall F1-scores. In this case, one should consider examples where both models give the right prediction, both give wrong predictions, MLP alone gives the right prediction, and FLAN-T5-small alone gives the right prediction. In such a way, diverse examples-based comparison is critical since the overall measure of performance fails to explain how the models work for each individual textual expression. The reason is that MLP is assumed to work properly when the texts contain some explicit word forms which statistically correlate with a certain class since these patterns are encoded by means of TF-IDF features. On the contrary, FLAN-T5-small can be more effective in examples which require context-based understanding of the relationship between words. Thus, incorrect predictions can be caused by the presence of ambiguous expressions, mixed sentiments, unusual wording, etc.

3.3 Comparative Analysis of Model Prediction

Output samples generated using the MLP model and the LLM-based Prompt 2 approach for the textual reviews selected from Data ID 75 are given in Table 4. The two sentiment classes are denoted using binary values 0 and 1 in this experiment. Examples illustrate that the two models classify similar sentiment in many cases, especially for reviews including quite obvious positive sentiment or good functioning of the product. For instance, one-word or one-phrase reviews like “Great product for under the sink” are always classified the same way by the two approaches. Nonetheless, discrepancies in classification are possible due to the different ways models work with text data. Thus, the MLP model uses numeric data and patterns found during training of the model, while Prompt 2 is based on the language understanding abilities of the LLM. Hence, in case of ambiguity or dependency of the

ID	Text	Model 1 (MLP)	Model 2 (Prompt 2)
75	Pretty Impressed I could fix my Maytag washer ...	1	0
75	Great product for under the sink	1	1
75	Replace your refrigerator filter every 6 months...	1	1
75	My husband YouTube'd how to fix my drier when ...	1	1
75	Part fit perfectly. You might have to fiddle with...	1	1

Table 4: Sample predictions generated by the MLP model and LLM-based Prompt 2.

meaning on context, the two models may classify the same review differently.

4 Task4: Discussion and Summary

4.1 Summary

For this work, the following methods were used for binary text classification: a supervised learning Multilayer Perceptron (MLP) and the zero-shot FLAN-T5-small language model. The MLP model was trained on TF-IDF representation of texts using unigrams and bigrams, achieving macro F1 score of 0.76. In turn, the FLAN-T5-small language model scored 0.848 F1 score using Prompt 2, thus performing better at classification. The results suggest that the context-aware representation of texts and ability to follow instructions by the FLAN-T5-small model were more advantageous than feature representation used by the MLP model. In order to explore the effect of the formulation of prompts, five different prompts were examined and the best one is Prompt 2. Still, the MLP model retains its significance as an efficient supervised baseline.

4.2 Lessons Learned

There were several lessons learned from conducting and evaluating both classification models. The first one is that correct text representation makes a big difference in model performance. While TF-IDF offers a quick and good numeric representation for the MLP model, it does not have great capacity to represent any kind of contextual and semantic information. The second lesson learned from the experiment was that zero-shot language models can do classification tasks without having any extra training, but they depend greatly on how the instructions for doing that are written. Experimenting with five different prompts showed that even small changes in prompt wording might result in different F1-scores, which shows how important it is to systematically evaluate prompts rather than choosing them randomly. Also, the process made us understand the value of having a validation set to evaluate model performance and choose the best

configuration before generating test results. Finally, analysing particular examples alongside with F1-scores gave us a better understanding of how the models behave.

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